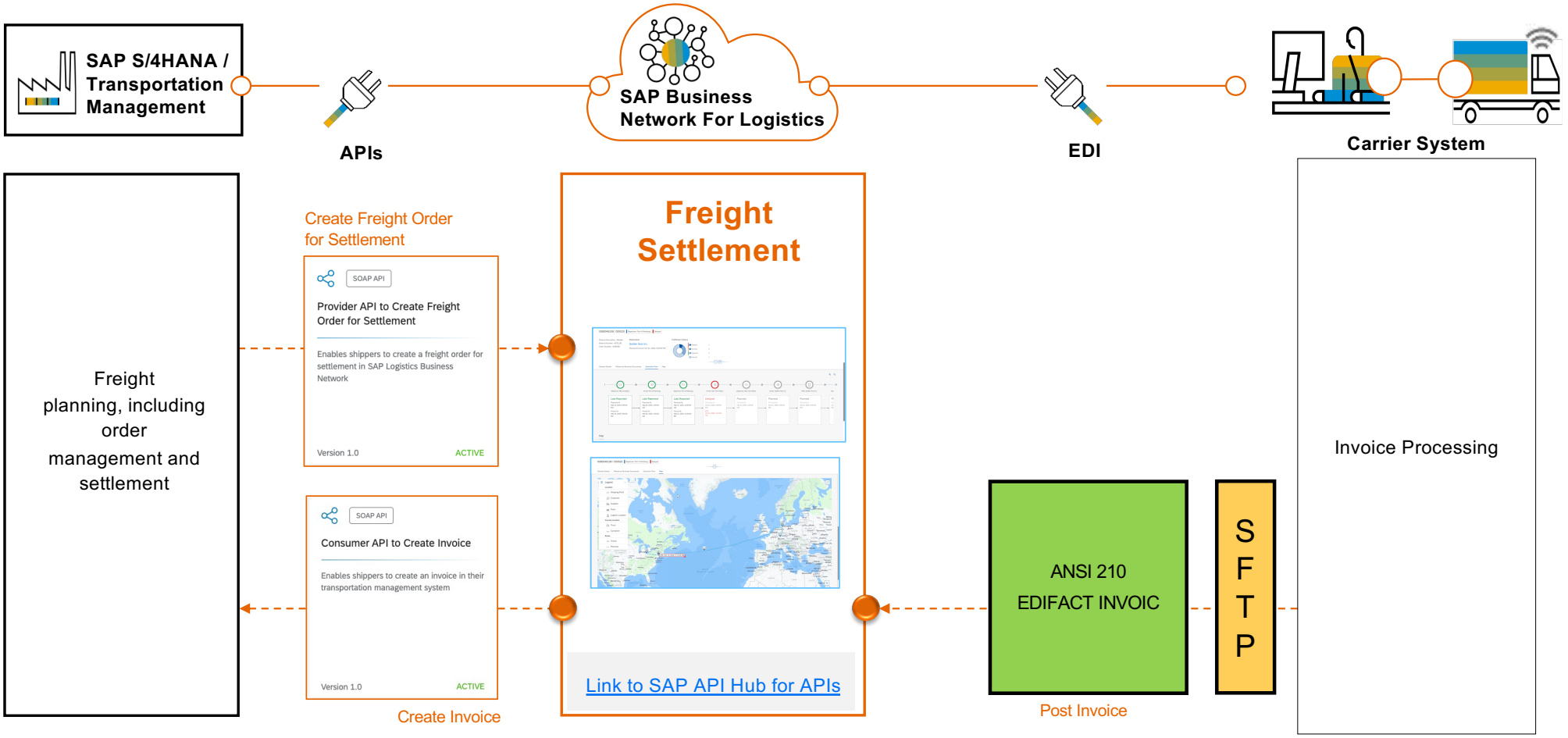
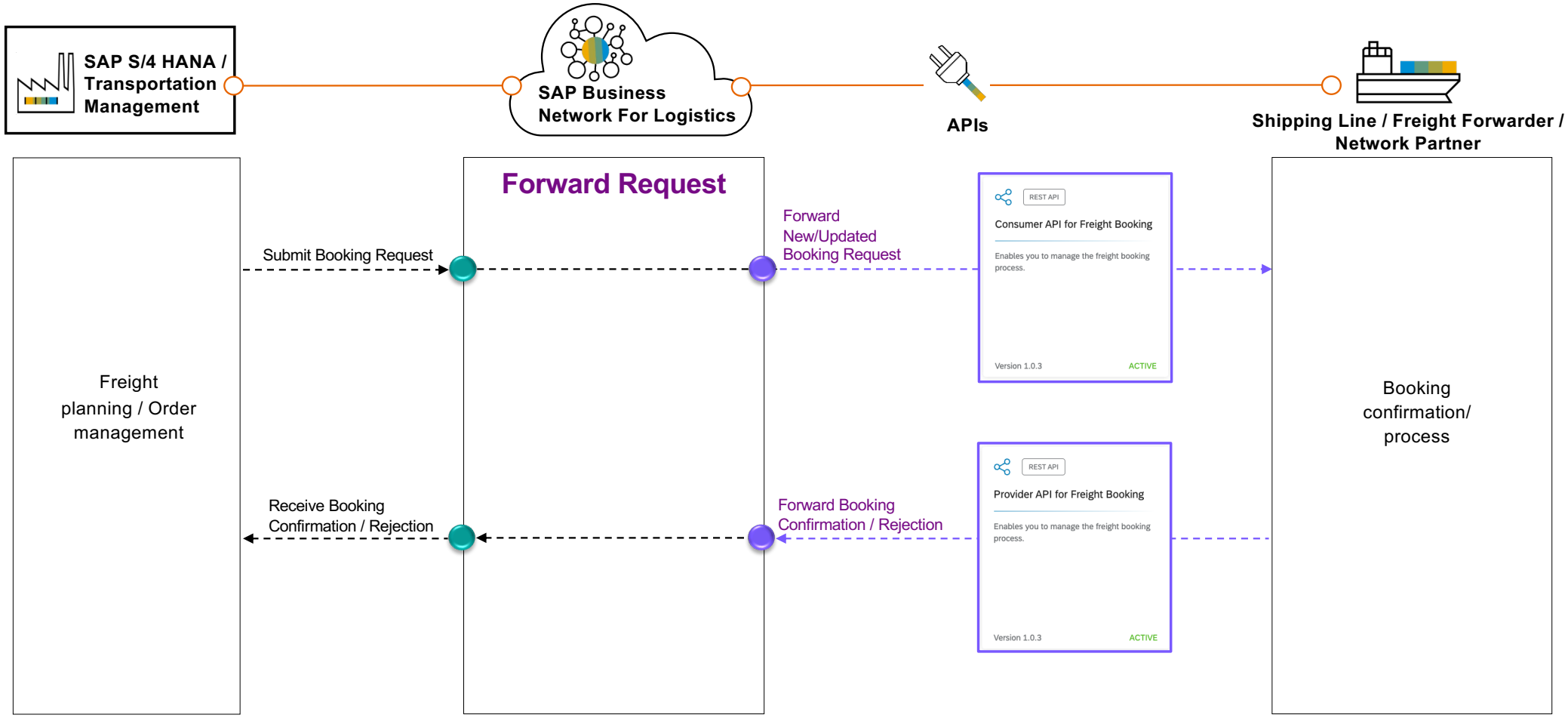
EDI for Invoicing



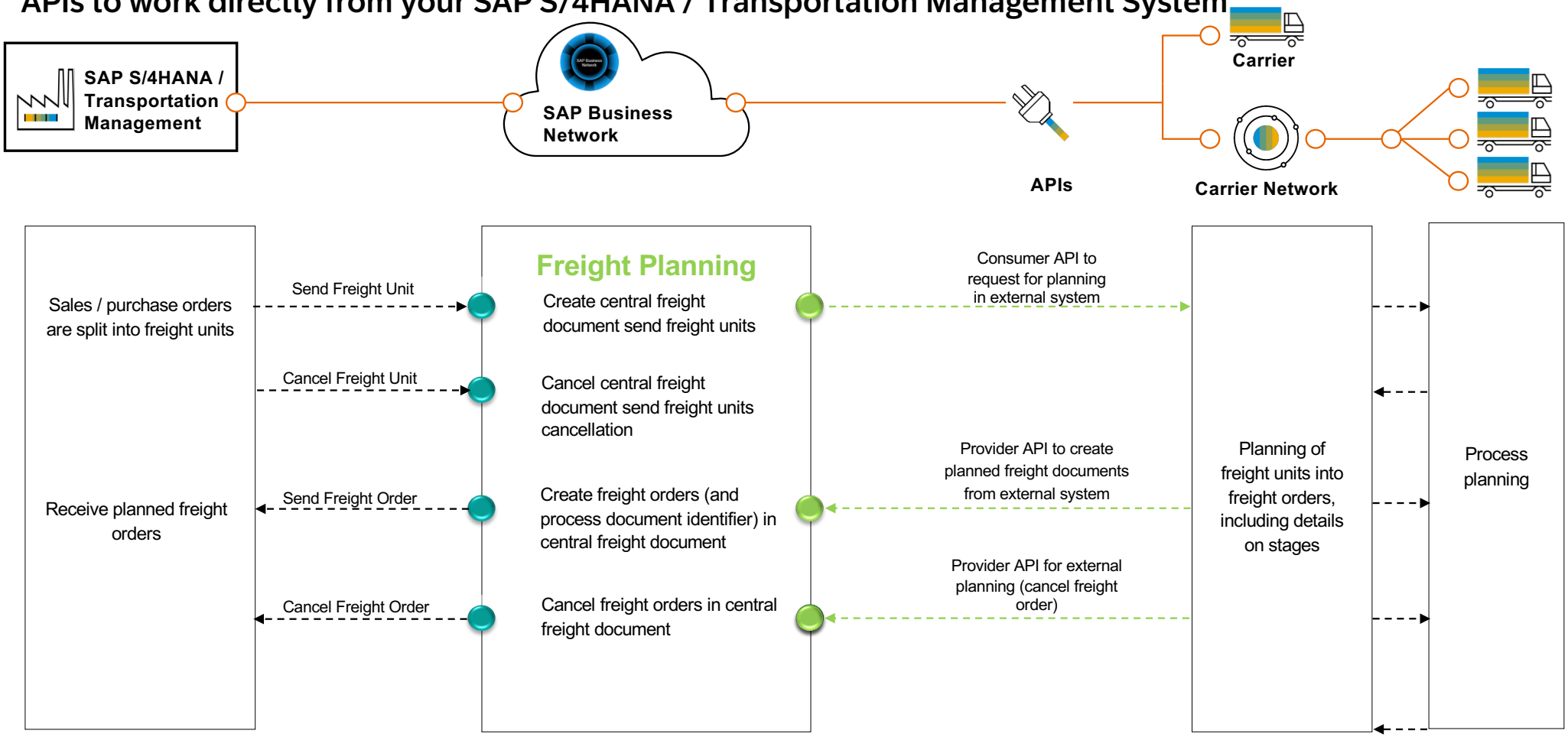
APIs for Ocean / Air Booking



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Leverage Freight Forwarders For External Planning

APIs to work directly from your SAP S/4HANA / Transportation Management System



Out-of-the-box integration via 'TransportationOrderGenericRequest'

Link to SAP API Hub for APIs

Carrier API / EDI Integration – What It Takes To Succeed

SAP S/4HANA / Transportation Management

- Every wrong step in a TM implementation will trickle down to BN4L
- Must implement TM with and understanding of how it works with BN4L

A Good SAP TM Implementation

Carrier Technical Expertise

- Carriers need to have the right technical resources – not all do
- Developers who can create and consume API / EDI and troubleshoot issues
- Carriers need to prioritize the integration development





Integration Process

1. Technical Kick-off Call
2. Integration Config Call
3. Regular Integration Testing Meetings



Close Shipper/Carrier/SAP Collaboration

Help from SAP (or SI)



- Extremely important that the two parties work together
- Processes need to be clear and understood by both shipper and carrier
- Alignment on test cases and their execution
- **Shipper project manages their carrier integrations**

- Must have someone from SAP assisting and troubleshooting
- Technical and business process expertise



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Customer Success Story – Large Pharmaceutical Company

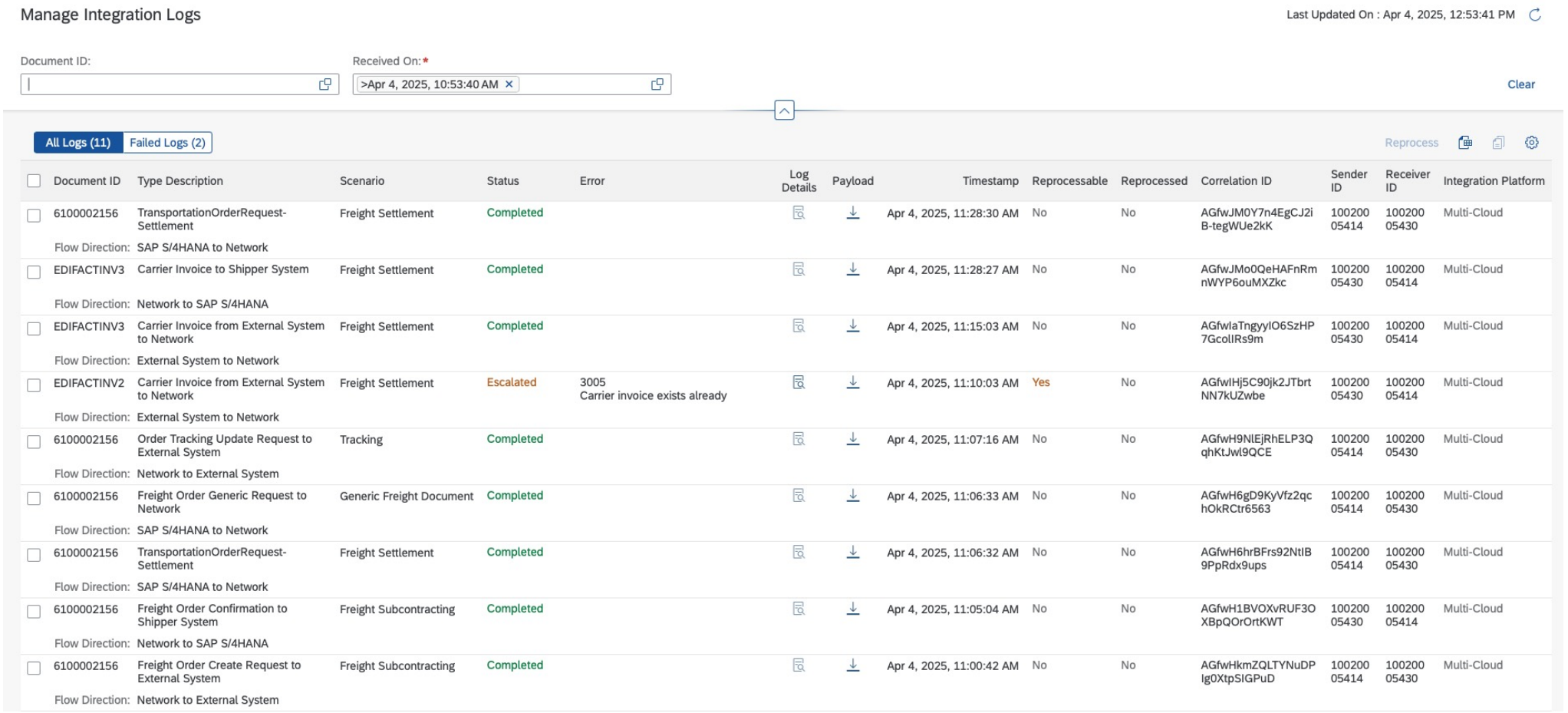
Tracking 100s of monthly door to door shipments globally in all modes (Road, Ocean & Air)

- Integrated 10+ Global Forwarders and Carriers
- Freight Booking and Freight Order integration (Confirmation and Order Events/Tracking)
- Started with Visibility Provider for tracking but abandoned
  - VP didn't support door to door
  - VP didn't support all regions they required
- Changed to direct integration to Freight Forwarders and Carriers
  - Advantages:
    - FFWs/Carriers support door to door tracking
    - FFWs/Carriers support many more events than VP
    - FFWs/Carriers support all regions
  - Disadvantages
    - No GeoLoc event support but wasn't business critical



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Before We Start – Manage Integration Logs App



Document ID	Type Description	Scenario	Status	Error	Log Details	Payload	Timestamp	Reprocessable	Reprocessed	Correlation ID	Sender ID	Receiver ID	Integration Platform
6100002156	TransportationOrderRequest-Settlement	Freight Settlement	Completed				Apr 4, 2025, 11:28:30 AM	No	No	AGfwJM0Y7n4EgCJ2iB-tegWUe2kK	10020005414	10020005430	Multi-Cloud
Flow Direction:	SAP S/4HANA to Network												
EDIFACTINV3	Carrier Invoice to Shipper System	Freight Settlement	Completed				Apr 4, 2025, 11:28:27 AM	No	No	AGfwJMo0QeHAFnRmnWYP6ouMXZkc	10020005430	10020005414	Multi-Cloud
Flow Direction:	Network to SAP S/4HANA												
EDIFACTINV3	Carrier Invoice from External System to Network	Freight Settlement	Completed				Apr 4, 2025, 11:15:03 AM	No	No	AGfwIaTngyyIO6SzHP7GcolIRs9m	10020005430	10020005414	Multi-Cloud
Flow Direction:	External System to Network												
EDIFACTINV2	Carrier Invoice from External System to Network	Freight Settlement	Escalated	3005 Carrier invoice exists already			Apr 4, 2025, 11:10:03 AM	Yes	No	AGfwIHj5C90jk2JTbrtNN7kUZwbe	10020005430	10020005414	Multi-Cloud
Flow Direction:	External System to Network												
6100002156	Order Tracking Update Request to External System	Tracking	Completed				Apr 4, 2025, 11:07:16 AM	No	No	AGfwH9NlEjRhELP3QqhKtJwl9QCE	10020005414	10020005430	Multi-Cloud
Flow Direction:	Network to External System												
6100002156	Freight Order Generic Request to Network	Generic Freight Document	Completed				Apr 4, 2025, 11:06:33 AM	No	No	AGfwH6gD9KyVfz2qchOkRCtr6563	10020005414	10020005430	Multi-Cloud
Flow Direction:	SAP S/4HANA to Network												
6100002156	TransportationOrderRequest-Settlement	Freight Settlement	Completed				Apr 4, 2025, 11:06:32 AM	No	No	AGfwH6hrBFrs92NtlB9PpRdx9ups	10020005414	10020005430	Multi-Cloud
Flow Direction:	SAP S/4HANA to Network												
6100002156	Freight Order Confirmation to Shipper System	Freight Subcontracting	Completed				Apr 4, 2025, 11:05:04 AM	No	No	AGfwH1BVOXvRUF3OXBpQOrOrtKWT	10020005430	10020005414	Multi-Cloud
Flow Direction:	Network to SAP S/4HANA												
6100002156	Freight Order Create Request to External System	Freight Subcontracting	Completed				Apr 4, 2025, 11:00:42 AM	No	No	AGfwHkmZQLTYNuDPIg0XtpSIGPuD	10020005414	10020005430	Multi-Cloud
Flow Direction:	Network to External System												

Agenda

Carrier Integration

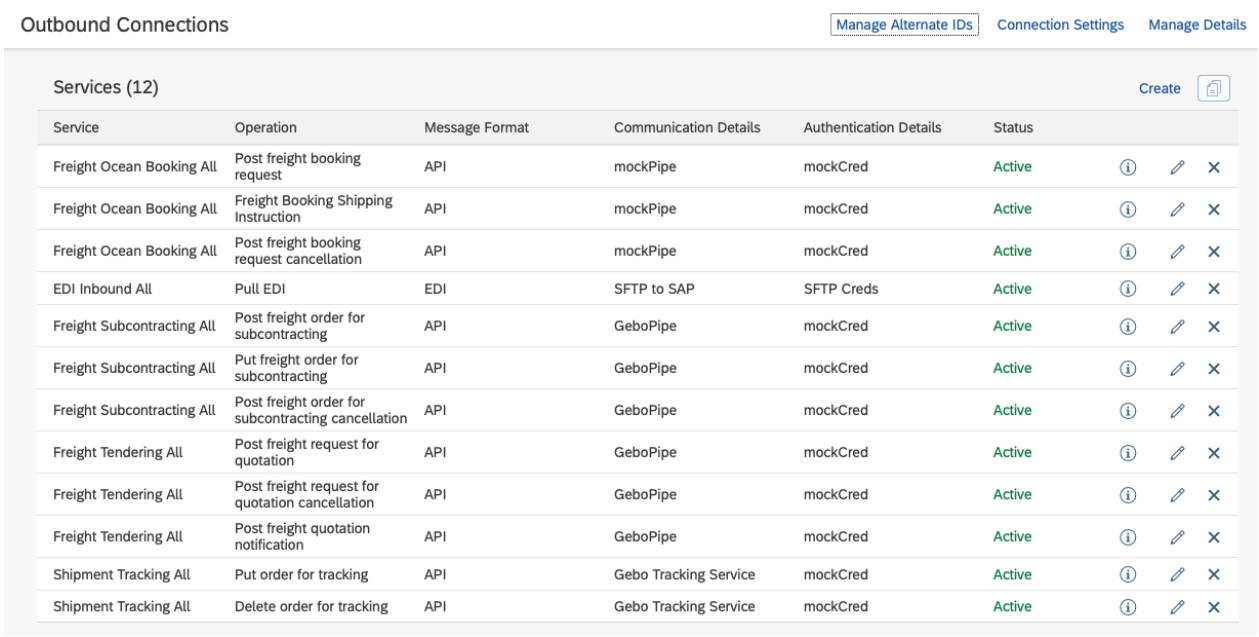
- Integration Overview
- **API Integration Details**
- EDI ANSI X12 Integration
- EDI EDIFACT Integration
- Odds & Ends

Day 3

System Connections – API Configuration

System Connections app is one stop shop for setting up carrier side integrations

- Main screen – define services you support (can also test on a test tenant)
- Connection Settings – define integration platform (always pick Multi-Cloud), and enable inbound connections
- Manage Details – define logical API endpoints
- Manage Alternate IDs – see your tenants LBN ID, enter in alternate IDs like DUNS, SCAC, GLN, USDOT, etc
  – Not needed for carrier integrations



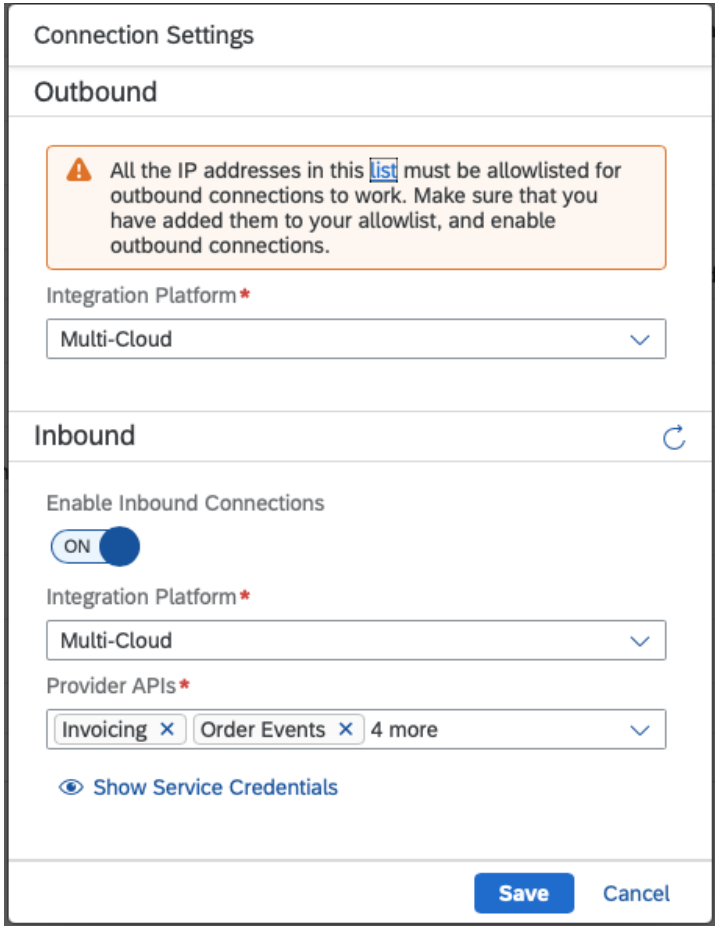
Service	Operation	Message Format	Communication Details	Authentication Details	Status
Freight Ocean Booking All	Post freight booking request	API	mockPipe	mockCred	Active
Freight Ocean Booking All	Freight Booking Shipping Instruction	API	mockPipe	mockCred	Active
Freight Ocean Booking All	Post freight booking request cancellation	API	mockPipe	mockCred	Active
EDI Inbound All	Pull EDI	EDI	SFTP to SAP	SFTP Creds	Active
Freight Subcontracting All	Post freight order for subcontracting	API	GeboPipe	mockCred	Active
Freight Subcontracting All	Put freight order for subcontracting	API	GeboPipe	mockCred	Active
Freight Subcontracting All	Post freight order for subcontracting cancellation	API	GeboPipe	mockCred	Active
Freight Tendering All	Post freight request for quotation	API	GeboPipe	mockCred	Active
Freight Tendering All	Post freight request for quotation cancellation	API	GeboPipe	mockCred	Active
Freight Tendering All	Post freight quotation notification	API	GeboPipe	mockCred	Active
Shipment Tracking All	Put order for tracking	API	Gebo Tracking Service	mockCred	Active
Shipment Tracking All	Delete order for tracking	API	Gebo Tracking Service	mockCred	Active

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System Connections > Connection Settings – API Configuration

System Connections > Connection Settings where you configure integration platform and turn on inbound connectivity and access OAuth details

- Integration Platform – **always pick Multi-Cloud** – Neo is deprecated. **Note**: if you know carriers still on Neo they need to migrate by mid May 2025!
- Inbound– turn on the switch if you want to call provider APIs
  - Select Multi-Cloud for integration platform
  - Select all provider APIs
  - Save and then come back to get Service Credentials (takes a few minutes to generate)
  - Service Credentials are OAuth Credentials, OAuth URL and API hostname – copy and use to call Provider APIs
  - IP addresses that carrier must allowlist if they have a firewall



System Connections > Manage Details – API Configuration

System Connections > Manage Details where you define Webhook Endpoints and Auth Details for those endpoints

- Communication Details – webhook endpoints to be used for consumer API service config
- Authentication Details – used to auth against webhook endpoints
  – Basic, API Key and OAuth 2.0 supported

